SEPARATION PROPERTIES OF THE SOURCE OF A MORPHISM WITH SEPARATED BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement lists six assertions in one enumeration; the proof treats them in two groups and the reduction uses the base Spec(Z).

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Stability of the separation axioms under composition. Lemma lemma-composition-separated (581-585), used at proof lines 623-626. Gives parts (4), (5) and (6).
- Separatedness of an algebraic space over a scheme in terms of the structure morphism. Spaces, Definition spaces-definition-separated and Properties of Spaces, Definition spaces-properties-definition-separated, used at proof lines 623-629. Identifies the absolute and relative forms of the three properties.
- The base Spec(Z) and its universal role. used at proof lines 627-629. Allows the relative case to be applied to the absolute one.

Source reference. Original lemma 03KQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4686–4686. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 02X5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3828–3828. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 03BS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4359–4359. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S.*

(1) *If Y is separated and f is separated, then X is separated.*
(2) *If Y is quasi-separated and f is quasi-separated, then X is quasi-separated.*

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(3) *If Y is locally separated and f is locally separated, then X is locally separated.*
(4) *If Y is separated over S and f is separated, then X is separated over S.*
(5) *If Y is quasi-separated over S and f is quasi-separated, then X is quasi-separated over S.*
(6) *If Y is locally separated over S and f is locally separated, then X is locally separated over S.*

Proof. Parts (4), (5), and (6) follow immediately from Lemma 03KQ and Spaces, Definition 02X5. Parts (1), (2), and (3) reduce to parts (4), (5), and (6) by thinking of X and Y as algebraic spaces over $\mathrm{Spec}(\mathbf{Z})$, see Properties of Spaces, Definition 03BS. □